

Furqan Asif

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Teaching Assistant

University of Engineering and Technology

Sep 2024 – July 2025

Lahore, Punjab

Programming Fundamentals (PF)

- Conducted tutorial sessions for **30+ students**, covering control structures, functions, arrays, and problem-solving in C/C++.
- Provided hands-on coding assistance during labs, helping students debug and understand core programming logic.
- Collaborated with faculty to refine course delivery, contributing to improved student engagement and performance.

Object-Oriented Programming (OOP)

- Assisted **10+ students** in understanding classes, objects, inheritance, and polymorphism through structured C++ exercises.
- Led practice sessions reinforcing OOP principles with real implementation tasks in C++.
- Worked with course instructor to align session content with exam objectives, improving overall conceptual clarity.

Education

University of Engineering and Technology, Lahore

Bachelor of Science in Computer Science (CGPA: 3.40 / 4.00)

2023-2027

Lahore, Punjab

Skills

Languages: C#, C++, Python, SQL, JavaScript, HTML, CSS, Bash

Frameworks & Libraries: .NET (ASP.NET), Node.js, React.js, Next.js, Nest.js, Streamlit, OpenCV, TensorFlow

Databases: PostgreSQL, MongoDB, NoSQL Databases

AI / ML: Machine Learning, Deep Learning, CNNs, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Model Evaluation

Technologies & Tools: REST APIs, Git/GitHub, Linux, Docker (if used), Bash Scripting, Microservices Concepts

Projects

Receptrib: AI-Powered Receptionist | 🔗

AI, NLP, LLM, Automation, API Integration

- Built an **AI receptionist system** automating inquiry handling, call routing, and appointment scheduling for **3+ client businesses**, reducing manual front-desk workload by **60%**
- Designed an **NLP-based intent classification pipeline** processing **100+ daily interactions** with autonomous request handling and response generation
- Integrated **calendar and communication APIs** enabling real-time scheduling synchronization across **multiple client channels**

MindVault: AI-Powered Second Brain | 🔗

Python, RAG, LangChain, Vector Search, Embeddings, AI Agents

- Developed a **RAG-based knowledge management system** enabling natural language querying over personal knowledge bases with context-aware AI responses
- Implemented **embedding-based semantic search** using vector databases, achieving **sub-second retrieval** from large unstructured datasets
- Built modular **AI agent workflows** supporting scalable knowledge ingestion and intelligent multi-turn conversational retrieval

AI Face Mask Detection | 🔗

Python, OpenCV, CNN, Deep Learning, Computer Vision

- Built a **real-time face mask detection system** using **CNN architecture**, achieving **90%+ classification accuracy** on test datasets

- Designed an end-to-end **computer vision pipeline** covering data preprocessing, model training, evaluation, and **live inference via OpenCV**
- Optimized model for **real-time inference** on video streams, demonstrating applied **deep learning and edge deployment** concepts

AFC-MAD: Android Food Ordering App | 

Kotlin, Android Studio, XML, Mobile Development

- Developed a **multi-screen Android application** in Kotlin with **5+ screens** simulating real-world food ordering and e-commerce workflows
- Implemented **UI navigation, state handling, and activity lifecycle management** for a smooth end-to-end user experience
- Delivered a fully functional **Android prototype** covering frontend design through application testing and deployment

CustomDNS: DNS Filtering System | 

Python, Networking, DNS Protocol, Systems Programming, Bash

- Built a **custom DNS server in Python** intercepting and processing domain resolution requests with **selective hostname-to-IP control**
- Implemented **DNS-level domain blocking** restricting access to targeted domains at the **network layer** without third-party tools
- Tested across a **local network environment** with **Bash automation scripts** for deployment, execution, and DNS behavior validation

CustomOS: GUI-Based OS Simulator | 

C, GTK, Operating Systems, Process Scheduling, Multitasking

- Developed a **Linux-based OS simulator in C** with **user/kernel mode separation**, system calls, and low-level control flow
- Implemented **preemptive multitasking and process scheduling algorithms** simulating core OS kernel behavior across **multiple concurrent processes**
- Designed a **GTK-based GUI shell** with an interactive system launcher, demonstrating full-stack **systems programming** from kernel logic to UI

Achievements & Certifications

SQL Basic, Intermediate and Advance Hacker Rank Certifications	Mar 2025
Learn React : Basic and Advance, Learn JavaScript Codecademy Certifications	April 2025
Kickoff Agile Project Management, Predictive Project Management PMI (Project Management Institute)	Mar 2025
Lead Manager ITEC G2G Department of a Computer Science, UET Lahore	Jan 2025 - Feb 2025
Captain, University Football Team UET, Lahore	Jan 2023 - Present
Teacher Assistant Department of a Computer Science, UET Lahore	Dec 2024 - Mar 2025
Star Performer of the Year in PF and OOP Department of a Computer Science, Uet Lahore	Sep 2024